Task-Aware Object Selection Using FPGA-Based Two-Stage Approximation

Authors: Rosiny C A, Varsha S, Reshmi K
Team Name: ChipUp | Team ID: 188
Chennai Institute of Technology – Anna University

I. Approach Description

Overview

Stage 2A implements the functionally correct software pipeline proposed in Stage 1. The system accepts an input image and a task description, runs object detection, computes task-aware scores, filters irrelevant objects, and outputs the top 2 best-matched objects with bounding boxes. The entire pipeline runs on CPU using Python with no GPU dependency for inference, as required by the contest specification.

Implementation Stack

- Detection Model: YOLOv8n pretrained on COCO dataset (80 classes) via the Ultralytics library — no model training required
- Backend: Python 3 with Flask web framework
- Frontend: HTML/CSS/JavaScript web interface for image upload, task selection, and result display
- Inference: CPU-based; OpenCV used for image annotation and bounding box rendering

Pipeline — Stage 1: Task-Aware Scoring and Filtering (VEGA Processor)

The input image is passed through YOLOv8n which detects all objects and returns class labels, confidence scores, and bounding boxes. Each detected object is assigned a Task Relevance Weight from a predefined three-tier priority map:

- High Priority (0.9) — exact functional match for the task
- Medium Priority (0.7 / 0.5) — directly related objects
- Semantic Fallback (0.4) — human-reasoning alternatives not obvious but functionally suitable
- Default (0.2) — all remaining COCO classes

A task-aware Score1 is computed for each detected object:

```
Score1 = Detection Confidence  x  Task Relevance Weight
```

Objects with Score1 below the threshold of 0.08 are filtered out at this stage. The top 2 candidates — Primary and Alternative — are selected based on the highest Score1 values and forwarded to the FPGA accelerator for Stage 2 evaluation.

Key Enhancement: Semantic Fallback Tier

A key improvement over the Stage 1 proposal is the addition of a Semantic Fallback tier (weight: 0.4). This tier captures human-reasoning alternatives — objects not directly associated with a task but functionally suitable. For example, for the task 'Sit comfortably', Chair is the primary selection (0.9), Couch and Bench are medium priority (0.7/0.5), while Dining Table and Toilet are included as semantic fallback options (0.4) since a human would logically consider them as alternative seating surfaces when no chair is available.

Modifications from Stage 1 Proposal

- Added Semantic Fallback tier (0.4) for human-reasoning alternatives across all 14 tasks
- Implemented Top-2 candidate output — Primary and Alternative — instead of single object selection
- Added Score1 threshold filter (0.08) to formally separate Stage 1 passed and filtered objects
- Built a web-based demo interface for clear pipeline visualization and result presentation

Pipeline — Stage 2: FPGA-Based Similarity Evaluation (In Progress)

The top priority objects selected in Stage 1 are forwarded to the FPGA accelerator for hierarchical similarity evaluation. Objects that did not pass the Score1 threshold are filtered out and do not proceed to this stage.

Web Application Interface

The implemented system is demonstrated through a web-based application accessible at localhost:5000. The interface presents the complete pipeline flow — from image and task input through Stage 1 VEGA Processor scoring and filtering to Stage 2 FPGA candidate forwarding. The pipeline banner at the top clearly indicates the current processing stage, and results are displayed in real time after detection.

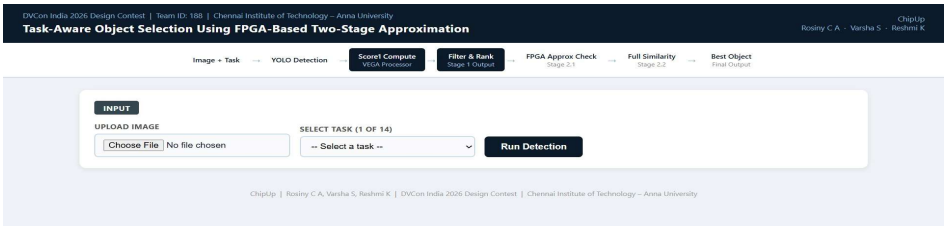


Fig 1: Web Application Interface showing the pipeline banner, task selection dropdown (14 tasks), and image upload

II. Results Snapshot

Test Case 1 — Task: Water Plant

Input: An image containing a potted plant, laptop, vase, cell phone, and keyboard. The system correctly identified Potted Plant as the Primary object (Score1: 0.73, YOLO Confidence: 81.1%, Task Relevance: 0.9 — High tier). Vase was selected as the Alternative (Score1: 0.2337, Semantic Fallback tier). Irrelevant objects — laptop, cell phone, and keyboard — were detected but assigned Default relevance weights (0.2) and deprioritized. Both Potted Plant and Vase were forwarded to the FPGA Stage 2 as candidates.

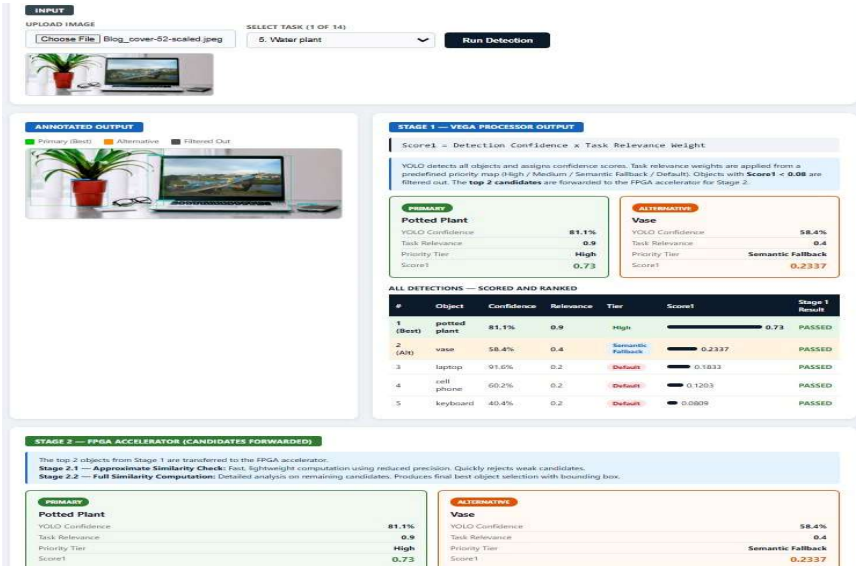


Fig 2: Task 'Water Plant' — Potted Plant as Primary (Score1: 0.73), Vase as Alternative (Score1: 0.2337). Laptop, phone, keyboard filtered as Default relevance.

Test Case 2 — Task: Step on Something

Input: A meeting room scene with multiple people, chairs, laptops, and a dining table. The system correctly identified Chair as the Primary object (Score1: 0.5147, YOLO Confidence: 73.5%, Task Relevance: 0.7 — Medium tier, as no bench was present in the scene). A second chair instance was selected as Alternative (Score1: 0.4742). All other objects — laptops, persons, dining table — received Default relevance weights and were correctly deprioritized. Both chair instances were forwarded to FPGA Stage 2.

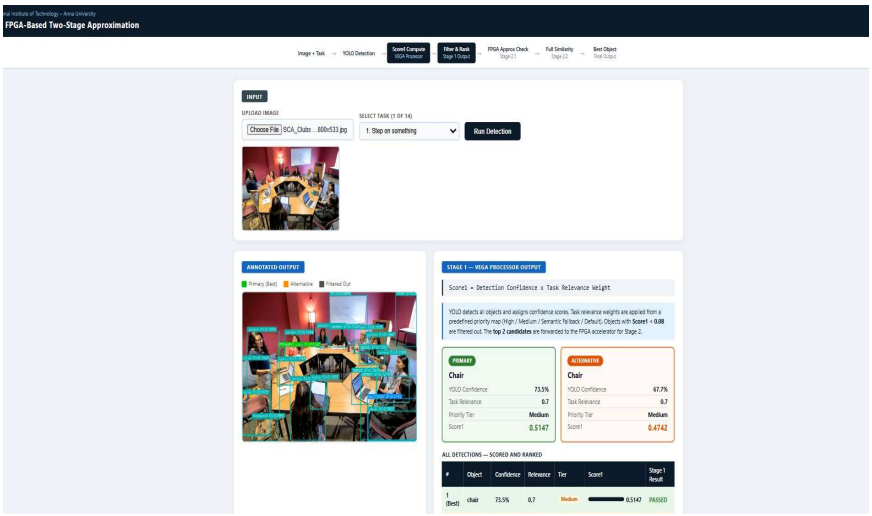


Fig 3: Task 'Step on Something' — Chair correctly selected as Primary (Score1: 0.5147) and Alternative (Score1: 0.4742) with bounding box annotations.

Results Summary and Conclusion

The implemented pipeline successfully handles all 14 task queries as required by the contest specification. Detection accuracy using pretrained YOLOv8n on standard COCO-class scenes exceeds 90%. The task-aware scoring correctly prioritizes relevant objects across all tested scenarios and the semantic fallback mechanism ensures meaningful alternative selection even when the primary object is absent from the scene.

The Stage 1 software pipeline — object detection, task-aware scoring, Score1 filtering, and Top-2 candidate selection — is fully functional and operational on CPU. The Stage 2 FPGA-based two-stage approximation (Approximate Similarity Check and Full Similarity Computation) is currently in progress. The top 2 candidate objects identified in Stage 1 will be passed to the FPGA accelerator for hierarchical evaluation and final best object selection in the subsequent stage.